

7.7 – Roots and Factors

Name: _____

Day 1: The Connection

Period: _____

TEKS 2A.7D: Determine the linear factors of a polynomial function of degree three and four using algebraic methods.

Four Names, One Idea

ROOTS = ZEROS = X-INTERCEPTS = SOLUTIONS

These all mean the same thing: the x -values where $y = 0$ (where the graph crosses the x -axis)

Same Concept, Different Contexts:

Word	When You'll Hear It	Example
Roots	"Find the roots of the polynomial"	$x = 2$ is a root
Zeros	"Find the zeros of $f(x)$ "	$x = 2$ is a zero
X-Intercepts	"Where does the graph cross the x -axis?"	$(2, 0)$ is an x -intercept
Solutions	"Solve the equation $f(x) = 0$ "	$x = 2$ is a solution

The Factor Theorem

If r is a ROOT, then $(x - r)$ is a FACTOR

$\updownarrow$ *Works both ways!* $\updownarrow$

If $(x - r)$ is a FACTOR, then r is a ROOT

△ Watch the Signs! Root $x = 3 \rightarrow$ Factor $(x - 3)$ Root $x = -2 \rightarrow$ Factor $(x - (-2)) = (x + 2)$

From Roots to Factors

Example 1: A polynomial has roots $x = 1$, $x = 4$, and $x = -2$. Write it in factored form.

Root $x = 1 \rightarrow$ Factor _____

Root $x = 4 \rightarrow$ Factor _____

Root $x = -2 \rightarrow$ Factor _____

Factored form: $f(x) =$ _____

Example 2: A polynomial has roots $x = 0$, $x = 5$, and $x = -3$. Write it in factored form.

Factored form: $f(x) =$ _____

From Factors to Roots

Example 3: $f(x) = (x - 5)(x + 1)(x - 3)$. Find all the roots.

Factor $(x - 5) \rightarrow$ Root _____ Factor $(x + 1) \rightarrow$ Root _____ Factor $(x - 3) \rightarrow$ Root _____

Roots: $x =$ _____

Example 4: $g(x) = x(x + 4)(x - 2)$. Find all the roots.

Roots: $x =$ _____

Verifying Roots with Synthetic Division

Key Idea: If you divide by $(x - r)$ and get remainder $= 0$, then r IS a root!

Example 5: Is $x = 2$ a root of $f(x) = x^3 - 6x^2 + 12x - 6$?

Use synthetic division with $c = 2$:

$$\begin{array}{r|rrrr} +2 & 1 & -6 & 12 & -6 \\ \hline \end{array}$$

Remainder = _____ Is $x = 2$ a root? YES / NO

Example 6: Is $x = 3$ a root of $f(x) = x^3 - 6x^2 + 11x - 6$?

$$\begin{array}{r|rrrr} +3 & 1 & -6 & 11 & -6 \\ \hline \end{array}$$

Remainder = _____ Is $x = 3$ a root? YES / NO

Example 7: Is $x = -2$ a root of $g(x) = x^3 + 4x^2 + x - 6$?

$$\begin{array}{r|rrrr} -2 & 1 & 4 & 1 & -6 \\ \hline \end{array}$$

Remainder = _____ Is $x = -2$ a root? YES / NO

Example 8: Is $x = -5$ a root of $f(x) = x^3 + 5x^2 - 16x - 80$?

$$\begin{array}{r|rrrr} -5 & 1 & 5 & -16 & -80 \\ \hline \end{array}$$

Remainder = _____ Is $x = -5$ a root? YES / NO